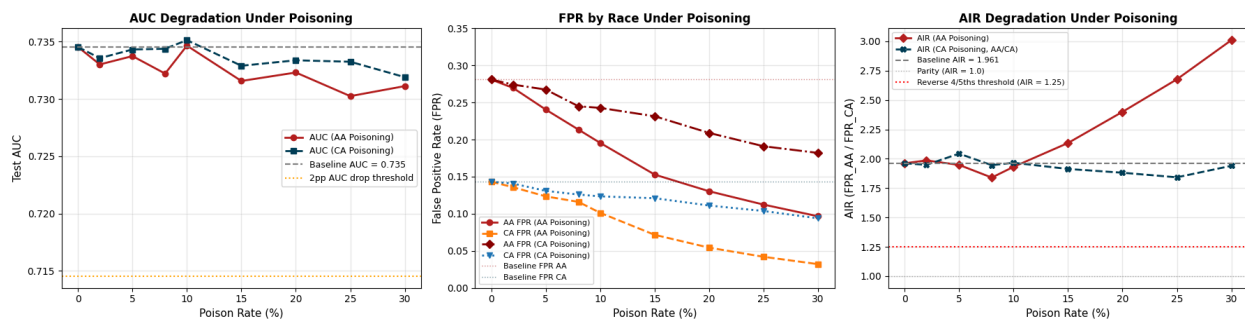


Security and Fairness Audit: COMPAS Logistic and GBT Model

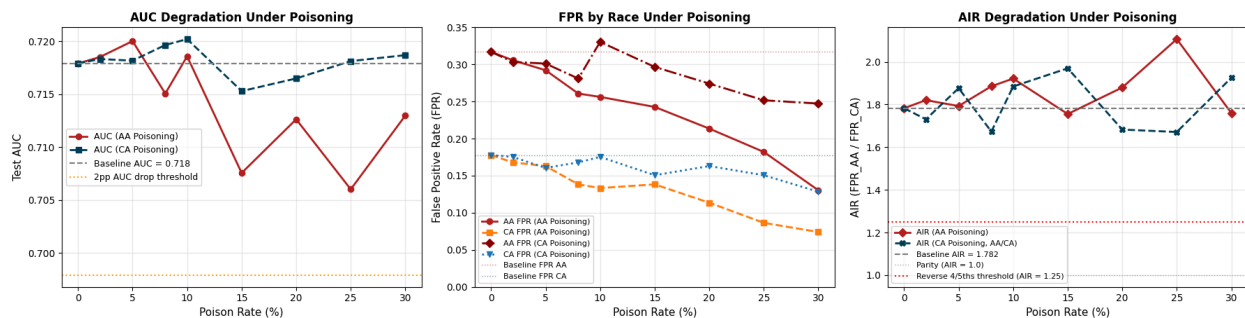
The Highest-Risk Finding: Stealth Zone Detected in Label-Flip Poisoning

This type of attack, assuming white-box data access to systemically underscoring Caucasian and African-American defendants and degrading AIR without triggering standard drift detection, reveals that both models are always above the inverse 4/5th rule threshold, making the stealth threshold at position rate of 0%. The AIR when poisoning African-American defendants (in the LR model) grows from 1.96 to nearly 3.0 at 30% poisoning. Meanwhile, the AUC decreases by 0.004, which is below the 2 pp stealth zone. This indicates that racial bias could be missed by model-performance monitoring (such as AUC). When poisoning Caucasian defendants, there isn't a strong increase in AIR, making it more of an effect targeted at the protected Black race.

Part B — Label-Flip Poisoning: AUC and Fairness Degradation (COMPAS · Logistic Regression)

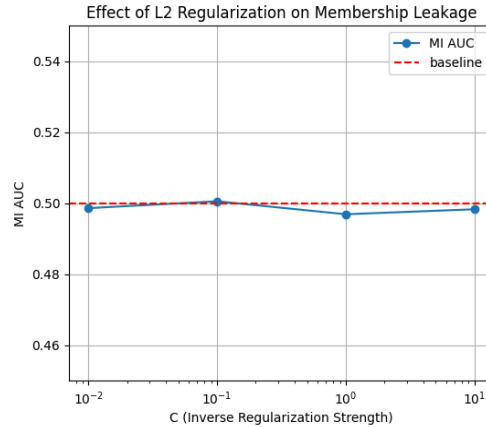


Part B — Label-Flip Poisoning: AUC and Fairness Degradation (COMPAS · GBT Model)



Proactive Mitigation:

The chart below shows that the model with L2 regularization is better defended against membership inference attacks. At $C = 1$, MI AUC remains at 0.497, showing that adding L2 regularization preserves minimal membership inference risk while introducing additional protection against overfitting, with no measurable increase in privacy leakage. While the red dashed line represents the MI AUC, in which an attacker is randomly guessing if the recording was in the training set, the membership inference attacks exploit a model's tendency to be more confident about the data it has seen and what it hasn't. By penalizing through L2 regularization, the model is further prevented from memorizing specific training points. The MI AUC remains near the 0.5 baseline across all levels of regulation, indicating its success in smoothing the decision boundary and making the model's behavior on training data indistinguishable from its behavior on non-training data.



Reactive Mitigation:

To prevent poisoning attacks from bypassing AUC-based drift detection, a more robust monitoring framework should track AIR for the African-American group, which provides an early signal of attack progression (rising from 1.98 at 2% poisoning to ~ 3.0 at 30% while AUC remains stable). When AIR falls outside the EEOC 4/5ths range (0.8–1.25), automated bias recalibration should adjust decision thresholds to restore parity while triggering root-cause analysis. This monitoring remains effective even under privacy constraints: at $\epsilon = 0.25$, logistic regression yields $\text{FPR_AA} = 0.569$ and $\text{FPR_CA} = 0.370$ ($\text{AIR} = 1.535$), showing disparities are still detectable and reinforcing the need for continuous group-level fairness checks.

Disparate Impact Tradeoffs:

The primary tradeoff of these mitigations is the push-and-pull between the model's utility and its privacy/fairness constraints. We see below that the LR model achieves better privacy by maintaining no gap between its training and non-training members (0.658 vs. 0.659). Yet, this is only done if the overall prediction confidence is sacrificed. The GBT model offers higher max predicted probabilities, but its wider gap between the training and test scores (0.702 vs. 0.697) creates more room for membership inference attacks to expose its memorization of the training data. Both the proactive and reactive measures mentioned above may lead to this tradeoff; there is more protection against membership inference risk, but the model has less predictive power. Reactive measures like the bias recalibration mentioned above may lead to a tradeoff. By shifting thresholds to comply with AIR, the model could experience a drop in accuracy or AUC.

Confidence Gap Histograms: LR vs. GBT Models

